

PRACTICAL-7

INPUT

```
GNU nano 8.7.1 memory.c
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>

int global_variable = 10;

void code_function()
{
    printf("This is a code function.\n");
}

int main()
{
    static int static_variable = 20;
    int stack_variable = 30;

    int *heap_variable = malloc(sizeof(int));
    *heap_variable = 40;

    printf("=====\n");
    printf("          PRACTICAL 7 - MEMORY LAYOUT\n");
    printf("=====\n");

    printf("Process ID          : %d\n", getpid());
    printf("Address of code function : %p\n", (void *)code_function);
    printf("Address of global variable: %p\n", (void *)&global_variable);
    printf("Address of static variable: %p\n", (void *)&static_variable);
    printf("Address of heap variable : %p\n", (void *)heap_variable);
    printf("Address of stack variable: %p\n", (void *)&stack_variable);

    printf("\nProcess is running...\n");
    printf("Use another terminal to inspect /proc/%d/maps\n", getpid());
    printf("Press Enter to terminate...\n");

    getchar();

    free(heap_variable);

    return 0;
}
```

Output

```
administrator@VBC0-LAP-29:~$ mkdir -p ~/practical-7
administrator@VBC0-LAP-29:~$ cd ~/practical-7
administrator@VBC0-LAP-29:~/practical-7$ pwd
/home/administrator/practical-7
administrator@VBC0-LAP-29:~/practical-7$ nano memory.c
administrator@VBC0-LAP-29:~/practical-7$ gcc memory.c -o memory
administrator@VBC0-LAP-29:~/practical-7$ ls
memory  memory.c
administrator@VBC0-LAP-29:~/practical-7$ ./memory
=====
PRACTICAL 7 - MEMORY LAYOUT
=====
Process ID           : 1810
Address of code function : 0x5b8f4588c209
Address of global variable: 0x5b8f4588f010
Address of static variable: 0x5b8f4588f014
Address of heap variable : 0x5b8f5c5d0010
Address of stack variable: 0x7ffe0aa5071c

Process is running...
Use another terminal to inspect /proc/1810/maps
Press Enter to terminate...

administrator@VBC0-LAP-29:~/practical-7$ nano memory.c
administrator@VBC0-LAP-29:~/practical-7$ |
```